# Syllabus

| Course code | IC606c (graduate) | Course name | Advanced Computer Architecture | credits | 3 |
|---|---|---|---|---|---|
| Instructor | Name : Hyokeun Lee | | Homepage : https://relacslab.github.io/ | | |
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| | Office hour : appointment is required | | | | |
| 1. Goals | Well defining an architecture is a key to achieve a robust and high-performance computing system. This lecture will take you to the road on various advanced technologies of computer architecture (e.g. out-of-order, prediction, coherence...). Although these technologies will be taught based on general-purpose processors, they have been ubiquitously extended in versatile platforms. | | | | |
| 2. Textbooks | **Main**: Patterson and Hennessy, "Computer Architecture: A Quantitative Approach"<br>**Sub**: (1) A. Gonzalez et al., "Processor Microarchitecture: An Implementation Perspective" (2) Research papers included in class slides | | | | |
| 3. Prerequisites | ▪ Computer Organization and Architecture (or similar class in undergraduates)<br>▪ C/C++ programming related classes | | | | |

| 4. Ratings (%) | Attendance | Homework | Mid-term | Final-term | Project | Others | Overall |
|---|---|---|---|---|---|---|---|
| | 0 | 0 | 50 | 50 | 0 | 0 | 100 |

| 5. Agenda | Week | Contents |
|---|---|---|
| | 1 | Introduction to Computer Architecture |
| | 2 | Recap of ISA and In-Order Microarchitectures |
| | 3 | Out-of-Order Microarchitecture (1) |
| | 4 | Out-of-Order Microarchitecture (2) |
| | 5 | Out-of-Order Microarchitecture (3) |
| | 6 | Branch Prediction |
| | 7 | Cache and Data Prefetch |
| | 8 | Mid-Term Exam |
| | 9 | Shared Memory: Cache Coherence (1) |
| | 10 | Shared Memory: Cache Coherence (2) |
| | 11 | Shared Memory: Memory Consistency (1) |
| | 12 | Shared Memory: Memory Consistency (2) |
| | 13 | Virtual Memory and Its Advanced Issues |
| | 14 | Hardware Technologies for Memory |
| | 15 | Final-Term Exam |

| 6. Notes for students | ▪ F will be given if cheating is caught no matter what case is<br>▪ One grade lower if not taking either mid-term or final-term exam |
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